

NOVA SOLVASON

Graduate Student – Rochester Institute of Technology
<https://novasolvason.github.io/> ♦ tas2557@rit.edu

EDUCATION

Rochester Institute of Technology (RIT)

Graduation: May 2027

Combined Accelerated BS/MS Program - Physics & Astrophysical Sciences and Technology

Minor in 3D Sculpture

Immersion in American Sign Language

Overall GPA: 3.840, Dean's List, Summa Cum Laude

TEACHING EXPERIENCE

Teaching Assistant - School of Physics and Astronomy

2023 - 2025

Rochester Institute of Technology

Rochester, NY

- University Physics I, University Physics II
- Responsibilities include assisting with lab setups, answering student questions in class, grading homework, and hosting a weekly recitation session
- 1-2 classes per semester
- Was awarded with the School of Physics and Astronomy's **Outstanding TA/LA Award** in Spring 2024

STEM Summer Camp Assistant

2023 - 2025

Challenge Island, Reno

Reno, NV

- Helping host week-long STEM based summer camps with activities that introduce kids age 6-12 into concepts of science and engineering
- Responsibilities include packing materials and setting up at the weekly location, managing activities, resolving any conflicts that arise, and monitoring the safety of the 40 children in attendance
- Similar experience at one-time events with organizations such mentoring younger students with STEM & Buds, Nevada, Keeping and eye on kids at the Olympic Valley Kids Camp, and tutoring at Roy Gomm Elementary School

PROFESSIONAL EXPERIENCE

Computer Lab IT Assistant

2025 - Present

Rochester Institute of Technology

Rochester, NY

- Assisting the staff and students at RIT with any technical issue that might arise
- Responsibilities include routine maintenance, management, and cleaning of all computers, printers, projectors, tablets, and their associated software present in the computer labs.
- Personally developed Standard Operating Procedure for backend testing of the *Canon Professional Print & Layout* software

Workshop Studio Monitor

2025 - Present

Rochester Institute of Technology

Rochester, NY

- Maintaining the machinery and tools within the sculpture studio workshop and assisted students as needed
- Responsibilities include weekly safety checkups on equipment, preparing and assisting with Foundry casting pours, helping students with the tools and machines, and taking any necessary action in the event of an emergency.

TECHNICAL EXPERIENCE

Programming Languages	Python, Java, C++, IDL, Scheme, TCL, LaTeX
Computer OS's	Windows, Mac, Linux, Embedded Virtual Machines, Remote Server Conections
Modeling Software	Solidworks, Autodesk Fusion 360, Sentaurs TCAD, MESA Stellar Evolution

RESEARCH

Pixel Design for Photon Counting IR Detectors	December 2025 - Present
<i>Center for Detectors (CfD) - Rochester Institute of Technology</i>	<i>Rochester, NY</i>

My research focuses primarily on astrophysics instrumentation. I have actively been working on the development and design of infrared detectors with incredibly low read noise and high conversion gain. The goal of this project is to design the pixel of a photon counting CMOS detector with a cutoff wavelength of $\lambda = 2.4\mu m$. I am planning to develop a Mercury-Cadmium-Telluride detector that has a read noise less than $0.5e^- R.M.S.$. This project falls under the scope of NASA ESI 2023 proposal by Don Figer, CfD.

As I have been learning the Synopsys Sentaurs TCAD Environment, I have developed a simple 3D Model of a Silicon pixel based on US Patent 11764249 by Jiaju Ma - CTO, Gigajot Technology. While I learn the software, I am developing a model and simulation code based on Ma's design such that I can then compare the results of my simulation with Ma's publications. Once I have developed and validated the results of this baseline simulation code, I will be converting the material of the detector from silicon to MCT ($Hg_{x-1}Cd_xTe$, with mole fraction $x = 0.502$). I will then move on to adjusting the design of the detector aiming to minimize the capacitance of the floating diffusion region and adjust the design as needed to increase conversion gain while keeping the read noise below the $0.5e^- R.M.S.$ threshold.

Investigating Intrinsic Patterns in Photon Counting CMOS Detectors	August 2025 - December 2025
<i>Center for Detectors (CfD) - Rochester Institute of Technology</i>	<i>Rochester, NY</i>

The focus of this project was to characterize several single photon sensing detectors in order to advance the technology to NASA TRL 5 for potential future use on the *Habitable Worlds Observatory*. The project is funded under the NASA SAT-2022, with PI Don Figer. While assisting Dr. Figer's research team at the Center for Detectors, I encountered a persistent banding pattern visible in the flat field calibration images taken with the Hamamatsu Quest camera and the BAE HWK-4123 detectors. My work on this project consisted of an investigation into the source of this banding pattern between alternating rows that was apparent in all of the flat-field images of both detectors. The results of this investigation leads me to conclude that this banding was caused by the hardware implemented to increase the readout speed of the camera. The Hamamatsu Quest and HWK 4123 have a synchronous dual-row readout feature using separate Analog to Digital Converter circuitry for alternating rows. The most likely cause of the banding pattern is from minor inconsistencies in the hardware leading to a difference in conversion gain. The difference in signal between the rows ranged from 0.5% – 1%, and given that it was an intrinsic pattern, it can be easily accounted for in the flat field calibrations of any science images as they are processed.

FELLOWSHIPS/GRANTS/AWARDS/HONORS

Outstanding TA/LA Award	<i>April 2024</i>
<i>Rochester Institute of Technology - School of Physics and Astronomy</i>	
RIT Presidential Scholarship	<i>August 2022 - December 2025</i>
<i>Rochester Institute of Technology - School of Physics and Astronomy</i>	
2nd Place Overall, Phoenix Award, Safety Award	<i>April 2022</i>
<i>NASA Human Exploration Rover Challenge</i>	

CONFERENCES

Global Physics Summit	March 2025
<i>American Physical Society</i>	<i>Anaheim, CA</i>

I was sponsored through the RIT School of Physics and Astronomy to attend the 2025 APS Global Physics Summit. Here I was able to attend talks about current research, go to networking events, and talk to representatives from companies and leading organizations at the front of groundbreaking research.

VOLUNTEERING/OUTREACH

Student Club Leadership - Founding Member and Vice President

March 2026 - Present

RIT Minorities in Physics and Astronomy

Rochester, NY

The RIT Minorities in Physics and Astronomy Club was established to provide a safe space and cultivate a positive community for the underrepresented groups in the RIT Physics and Astronomy programs. We seek to support students who may need it by providing resources and hosting meetings where members can share their experiences and where we can discuss how to make the field more equitable and inclusive.

- Responsibilities include event and meeting planning, room reservation, communication with RIT Faculty and outreach to students in the Physics and Astronomy programs at RIT.

Astronomy Club Outreach - Solar Observing Events

January 2026 - Present

RIT Astronomy Club

Rochester, NY

The RIT Astronomy Club hosts weekly (weather permitting) Solar Observing events on campus. These events consist of me and a group of graduate students setting up a solar telescope with an H_{α} filter and hosting an open event where students from all majors and backgrounds can safely look at and learn about our star.

- Responsibilities include helping to set up the solar telescope, sharing information about our sun, and answering and general questions attendees might have about astronomy.

Student Club Leadership - Vice President

April 2024 - April 2025

RIT Cosplay Troupe

Rochester, NY

The RIT Cosplay Troupe is the second largest club at RIT, and we aim to create a place where students can make friends and explore their creativity through the art of costume design. We hosted weekly meetings that included informational presentations on different crafts and techniques, hosted multiple dinner party events, put on two large scale stage productions, and ran the 2025 RIT SnowFest Cosplay Convention.

- Responsibilities include event planning, and coordination with the RIT administration, booking reservations for events, resolving conflicts that arise, and setting up fundraising events. I was also the production manager of both of our “Cosplay Chess” Shows, working closely with the directors of the show and taking care of most of the behind the scenes operations.